Black-hole spin-down with a self-interacting two-level axion cloud

Preliminary results, 10–11 November 2026

Catherine Welch

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Adiabatic multi-mode method, $|211\rangle + |322\rangle$, cosine potential, $\alpha_0 = 0.3$.
All results preliminary: one α, no GW emission, radial-mesh convergence not checked during evolution.

What is new: the black hole evolves

Cloud: occupations of the two phases change only through boundary fluxes (action balance),

$$\frac{d\mathcal{N}_i}{d\tau} = -L_{\mathcal{N}_i}, \qquad i \in \{211, 322\}.$$

Hole: absorbs what crosses the horizon,

$$\frac{dM}{d\tau} = \kappa\, F_E^{\mathrm{H}}, \quad \frac{dJ}{d\tau} = \kappa\, F_J^{\mathrm{H}}, \quad \kappa = (f_a/M_{\mathrm{Pl}})^2,$$

F^{H} = flux into the horizon in units of f_a^2; M_{Pl} non-reduced. Each step: nonlinear stationary radial solve on the current Kerr background.

Units: m_a only sets the scale. At $\alpha = 0.3$, $m_a = 10^{-12}$ eV $\leftrightarrow$ $M \approx 40\, M_\odot$, and $10^9 M_0 \approx 2.3$ days.

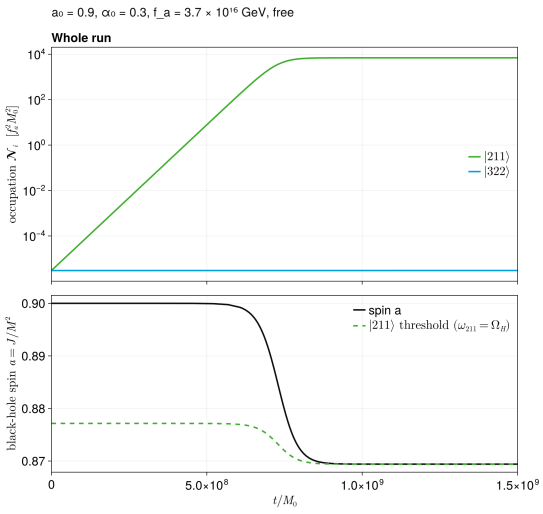
Runs (cosine unless marked):

a_0	f_a [GeV]	outcome
0.9	5.7×10^{18} (QCD)	free-like
0.9	5.7×10^{17} (QCD)	free-like, then $\|322\rangle$
0.9	3.7×10^{16}, free	reference
0.9	3.7×10^{16}	to $t = 10^{10} M_0$
0.99	3.7×10^{16}	unfinished at 1.5×10^9
0.99	1.2×10^{17}	timed out, 4.8×10^8

QCD: $m_a = 10^{-12}$, 10^{-11} eV ($M \approx 40$, $4\, M_\odot$).

Numerics fixed on day 1: warm-start every radial solve from the last *accepted* step; a failed solve rejects the step instead of being used.

Check: large f_a reproduces free-field spin-down

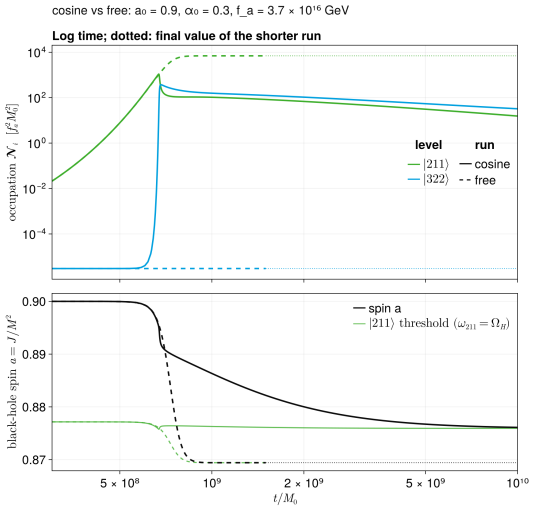


Spin locks onto the moving $|211\rangle$ threshold ($\omega_{211} = \Omega_H$).

run	final a	M/M_0	$\mathcal{N}_{211}$
free	0.86941	0.981478	6944
5.7×10^{18} GeV	0.86941	0.981478	0.2867
5.7×10^{17} GeV	0.86941	0.981475	28.68

- $\mathcal{N}$ in units of $f_a^2 M_0^2$: ratios 100.01 and 24219 match κ ratios.
- Time-shifted trajectories agree to $\leq 3 \times 10^{-5}$ in a.
- Lost J / lost $M = 3.4 = m/\omega_{211}$.

Self-interaction changes the path: $f_a = 3.7 \times 10^{16}$ GeV, $a_0 = 0.9$



	free	cosine
peak $\mathcal{N}_{211}$	6944	1086
final a	0.8694	0.8761
threshold	0.8694	0.8759
mass lost	1.85 %	0.30 %
J lost $[M_0^2]$	0.0625	0.0291
$\Delta J/\Delta M$	3.4	9.8

- $|211\rangle$ collapses at $\mathcal{N} \approx 1086$, $|322\rangle$ jumps by 10^8.
- Both end near their own threshold, but the cosine hole gets there losing half the J and 1/6 of the mass.
- Cosine run still approaching threshold at $10^{10} M_0$ (≈ 23 d at 40 $M_\odot$).

Why the self-interacting hole spins down more slowly

Spin-down rate $\approx$ (superradiant rate per quantum at this a) $\times\ \mathcal{N}_{211}$.

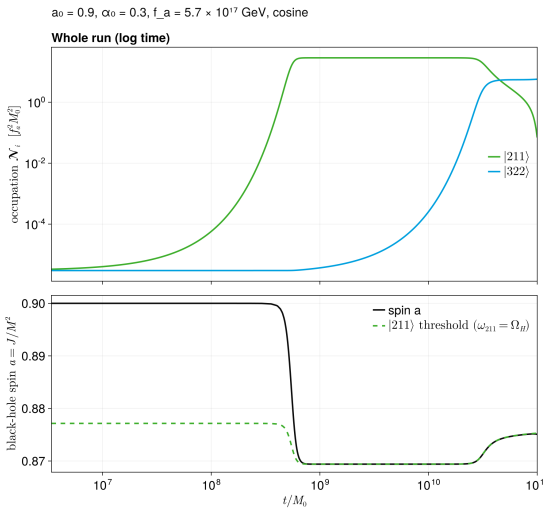
- Free: $|211\rangle$ grows until it holds all the available J; the spin falls while $\mathcal{N}_{211} \sim 10^3$–$10^4$.
- Cosine: transfer $2 \times |211\rangle \to |322\rangle$ caps $|211\rangle$ on a plateau, $\mathcal{N}_{211} \sim 100$ (superradiant gain = loss to $|322\rangle$).
- Plateau $\mathcal{N}$ in f_a^2 units is f_a-independent; free saturation $\propto (M_{\rm Pl}/f_a)^2$. Slow-down factor $\propto (M_{\rm Pl}/f_a)^2$, as in Baryakhtar et al. (2011.11646).
- Plateau shrinks as $a \to a_c$, so the spin creeps toward threshold.

Matched spin, after the cosine collapse (finite differences of accepted states):

a	$\dot{a}_{\rm free}/\dot{a}_{\rm cos}$	$\mathcal{N}_{211}^{\rm free}/\mathcal{N}_{211}^{\rm cos}$	$\mathcal{N}_{211}^{\rm cos}$
0.889	14	24	108
0.885	25	35	99
0.880	70	68	68
0.878	164	103	49

Rate tracks $\mathcal{N}_{211}$ to within $\approx 1.6\times$. Residual not separated: extra spin-down from $|322\rangle$ and the $m = 0$ channel at high a; different thresholds (different mass loss) near a_c.

Even QCD-sized f_a is not free at late times (5.7×10^{17} GeV, $4\,M_\odot$)



Restart of the saturated run to $10^{11} M_0$ (≈ 23 d at $4\,M_\odot$).

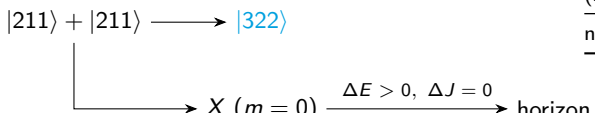
- $|322\rangle$ grows $200\times$ faster than free; excess rate $\propto \mathcal{N}_{211}^2$ (checked across the two QCD runs: ratio 10^4 for κ ratio 10^2).
- $|322\rangle$ stalls at $\mathcal{N} \approx 5.7$ (expected 50–100).
- $|211\rangle$ is reabsorbed ($\times 1/400$); hole *regains* mass, M/M_0: $0.9815 \to 0.9955$.
- Spin rises $0.8694 \to 0.8751$, tracking the threshold from just below.

At 5.7×10^{18} GeV the same excess is 2 % of the free rate.

Mechanism: the $m = 0$ component X is absorbed by the hole

Component $\mathbf{q} = (q_1, q_2)$ of the field: phase $q_1\xi_{211} + q_2\xi_{322}$ (ξ_i = phase of mode i, as in multiphase.tex), frequency $q_1\omega_{211} + q_2\omega_{322}$, $m = q_1 + 2q_2$. Call $X \equiv (2, -1)$:

$$\omega_X = 2\omega_{211} - \omega_{322} \approx 0.289 < \mu M, \qquad m_X = 0.$$



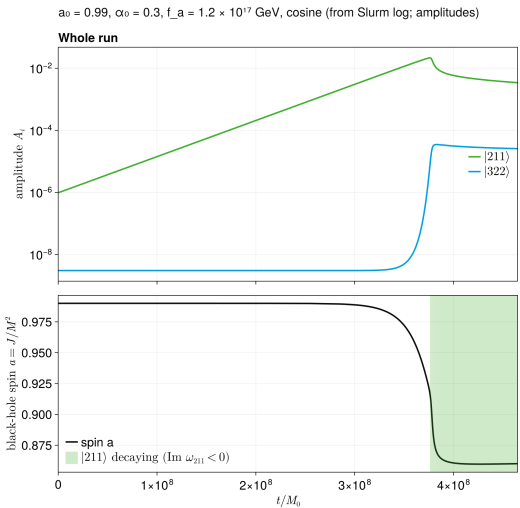
1. X cannot escape to infinity; at $m = 0$ it is never superradiant.
2. Energy without J lowers $a = J/M^2$ below the $|211\rangle$ threshold.
3. $|211\rangle$ is then absorbed, raising a; the mass gain raises α and the threshold.

Horizon flux by channel at $t = 2.9 \times 10^{10}$ (rebuilt solution):

channel	m	F_E^{H}	$\dot{a}$ part
$(1,0) = \|211\rangle$	1	$+2.3 \times 10^{-10}$	$+8.6 \times 10^{-13}$
$(0,1) = \|322\rangle$	2	-1.0×10^{-12}	-1.1×10^{-14}
$(2,-1) = X$	0	$+1.3 \times 10^{-10}$	-5.2×10^{-13}
net			$+3.3 \times 10^{-13}$

- Every channel has the sign of $\omega_c(\omega_c - m_c\Omega_H)$; channel sum = total flux to 5×10^{-26}.
- Threshold slope 0.415 per unit M/M_0 (free run) predicts final threshold 0.87521; recorded 0.87528.
- Not checked: is X resolved ($l = 0$ only)?

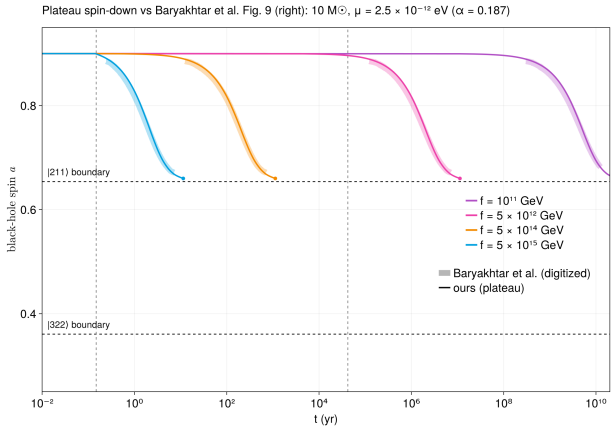
Competing regime (preliminary): $a_0 = 0.99$, $f_a = 1.2 \times 10^{17}$ GeV



- Peak mass loss 4.5 % ($M/M_0 = 0.955$ at 3.77×10^8).
- $|322\rangle$ grows at 7×10^{-7}, $4 \times 10^5 \times$ its linear rate, while $|211\rangle$ peaks.
- Spin crashes $0.918 \to 0.869$ in $\approx 7 \times 10^6 M_0$ and overshoots to 0.860, below the $|211\rangle$ threshold.
- Both levels then absorbed; mass recovers to 0.980.

Caveats: job timed out at 4.8×10^8 (no CSV; plot from log, amplitudes not occupations); mesh built at $a = 0.99$ used at 0.86; flux-sign test not run.

Quasi-static plateau spin-down vs Baryakhtar et al. Fig. 9



$10\,M_\odot$, $m_a = 2.5 \times 10^{-12}$ eV ($\alpha = 0.187$). For small f_a the hole moves $(M_{\rm Pl}/f_a)^2$ times slower than the plateau relaxes, so track the plateau:

$$\dot{a} = \kappa \left(F_J^{\rm H} - 2a F_E^{\rm H} \right) \equiv -\kappa\, D(a)$$

(M, α fixed; clock starts after free growth to the plateau, ≈ 0.13 yr).

- Against their digitized curves: our $t(a)$ is 6–9 % longer at $a = 0.78$, 2 % at $a = 0.70$, for every f.
- Near threshold ($a_c = 0.654$): $\mathcal{N}_{211} \propto (a - a_c)^{1/2}$, $D \propto (a - a_c)^{3/2}$, as in their equilibrium formulas.
- Free spin-down times: 0.149 yr ($|211\rangle$), 4.2×10^4 yr ($|322\rangle$, at $a = 0.654$) vs their 0.146, 5.0×10^4 (dashed lines).

Not yet: $f \geq 5 \times 10^{16}$ GeV and the $|322\rangle$ stage (need full evolution at $\alpha = 0.187$); mesh convergence.

Summary, open questions, next steps

Summary

- Coupled cloud + hole evolution runs end to end; large f_a reproduces free spin-down.
- Self-interaction caps $|211\rangle$: slower spin-down, $6\times$ less mass loss, larger $\Delta J/\Delta M$.
- The $m = 0$ partner of $2 \times |211\rangle \to |322\rangle$ feeds energy (no J) back into the hole; at late times this reabsorbs $|211\rangle$ even at QCD-sized f_a.
- Plateau spin-down matches Baryakhtar et al. Fig. 9 to $\lesssim 10\,\%$ in time.

Open / to discuss

- GW annihilation of $|211\rangle$: does it act on the 10^{10}–$10^{11} M_0$ times of the $m = 0$ effect?
- Is the $(2, -1)$ channel resolved with $l = 0$ only?
- Levels with $m \geq 3$ omitted.

Next

- Free + cosine pair at $\alpha = 0.187$ (direct test of the plateau method).
- Restart $a_0 = 0.99$ runs; containment check at low spin.
- Mesh-convergence check along one trajectory.